Zun Wang

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Zun's Google Scholar • wz0919

Education

University of North Carolina, Chapel Hill, Chapel Hill, North Carolina **Jul 2024 – Present**

Ph.D. in Computer Science
Advisor: Prof. Mohit Bansal

The Australian National University, Canberra, Australia **Jul 2020 – Jul 2023**

Master of Machine Learning and Computer Vision
Advisor: Prof. Stephen Gould
Graduated with Postgraduate Medal for Academic Excellence (**Top 1** out of the CECS college)

University of Science and Technology of China, Hefei, China **Sep 2015 – Jun 2019**

Bachelor of Mathematics and Applied Mathematics
Specialization in Probability and Statistics

Experience

Adobe Research, Research Intern May 2026 – Nov 2026
Topic: Real-time Multi-agent Video World Model with Persistent Shared Memory
Mentors: Yicong Hong, Yang Zhou, Chongjian Ge, Hao Tan

Amazon, Research Intern May 2025 – Nov 2025
Topic: Subject-Driven Video Generation and Evaluation (Accepted to NeurIPS 2026)
Mentors: Shan Yang, Kenan Deng

Research Interest

- Controllable Visual World Modeling
- Multimodal Understanding and Generation
- Multimodal Data Generation and Curation

Papers

* denotes equal contribution † Project Lead

1. **Beyond Copy-Paste: How Well Do Subject-Driven Video Models Understand Their Subjects? [Video Generation]**
   NeurIPS 2026
   Zun Wang, Kenan Deng, Daniel Blackburn, Linlin Lu, Junbang Liang, Yu Lou, Mohit Bansal, Shan Yang
2. **PhyMotion: Structured 3D Motion Reward for Physics-Grounded Human Video Generation [Video Generation]**
   NeurIPS 2026
   Yidong Huang*, **Zun Wang***, Han Lin, Dong-Ki Kim, Shayegan Omidshafiei, Jaehong Yoon, Jaemin Cho, Yue Zhang, Mohit Bansal
3. **AnchorWeave: World-Consistent Video Generation with Retrieved Local Spatial Memories [Video Generation][World Model]**
   ECCV 2026
   Zun Wang, Han Lin, Jaehong Yoon, Jaemin Cho, Yue Zhang, Mohit Bansal
4. **EPiC: Efficient Video Camera Control Learning with Precise Anchor-Video Guidance [Video Generation]**
   ICML 2026
   Zun Wang, Jaemin Cho, Jialu Li, Han Lin, Jaehong Yoon, Yue Zhang, Mohit Bansal

5. **DreamRunner: Fine-Grained Compositional Story-to-Video Generation with Retrieval-Augmented Motion Adaptation [Video Generation]**
   AAAI 2026
   Zun Wang, Jialu Li, Han Lin, Jaehong Yoon, Mohit Bansal
6. **Bootstrapping Language-Guided Navigation Learning with Self-Refining Data Flywheel [Embodied AI][Synthetic Data][Multimodal AI]**
   ICLR 2025
   Zun Wang, Jialu Li, Yicong Hong, Songze Li, Kunchang Li, Shoubin Yu, Yi Wang, Yu Qiao, Yali Wang, Mohit Bansal, Limin Wang
7. **Learning Goal-Oriented Language-Guided Navigation with Self-Improving Demonstrations at Scale [Embodied AI][Synthetic Data][Multimodal AI]**
   ECCV 2026
   Songze Li*, **Zun Wang*†**, Gengze Zhou, Jialu Li, Xiangyu Zeng, Limin Wang, Yu Qiao, Qi Wu, Mohit Bansal, Yi Wang
8. **Scaling Data Generation in Vision-and-Language Navigation [Embodied AI][Synthetic Data]**
   ICCV 2023 (***Oral***)
   Zun Wang*, Jialu Li*, Yicong Hong*, Yi Wang, Qi Wu, Mohit Bansal, Stephen Gould, Hao Tan, Yu Qiao
9. **Vision-and-Language Navigation Today and Tomorrow: A Survey in the Era of Foundation Models [Survey][Embodied AI][Multimodal AI]**
   TMLR 2024
   Yue Zhang*, Ziqiao Ma*, Jialu Li*, Yanyuan Qiao*, **Zun Wang***, Joyce Chai, Qi Wu, Mohit Bansal, Parisa Kordjamshidi
10. **Bridging the Gap Between Learning in Discrete and Continuous Environments for Vision-and-Language Navigation [Embodied AI]**
    CVPR 2022
    Yicong Hong*, **Zun Wang***, Qi Wu, Stephen Gould
11. **Planning with Sketch-Guided Verification for Physics-Aware Video Generation [Video Generation]**
    WACV 2027
    Yidong Huang, **Zun Wang**, Han Lin, Dong-Ki Kim, Shayegan Omidshafiei, Jaehong Yoon, Yue Zhang, Mohit Bansal
12. **Error-Driven Scene Editing for 3D Grounding in Large Language Models [Multimodal AI][Spatial Understanding]**
    ECCV 2026
    Yue Zhang, **Zun Wang**, Han Lin, Jialu Li, Jianing Yang, Yonatan Bitton, Idan Szpektor, Mohit Bansal
13. **Seeing Isn't Knowing: Do VLMs Know When Not to Answer Spatial Questions (and Why)? [Multimodal AI][Spatial Understanding]**
    Preprint, 2026
    Yue Zhang, **Zun Wang**, Han Lin, Yonatan Bitton, Idan Szpektor, Mohit Bansal
14. **V-Co: A Closer Look at Visual Representation Alignment via Co-Denoising [Visual Generation]**
    ECCV 2026
    Han Lin, Xichen Pan, **Zun Wang**, Yue Zhang, Chu Wang, Jaemin Cho, Mohit Bansal
15. **When and How Much to Imagine: Adaptive Test-Time Scaling with World Models for Visual Spatial Reasoning [Multimodal AI][Spatial Understanding]**
    NeurIPS 2026
    Shoubin Yu*, Yue Zhang*, **Zun Wang**, Jaehong Yoon, Huaxiu Yao, Mingyu Ding, Mohit Bansal

16. **Vision-and-Language Navigation with Analogical Textual Descriptions in LLMs** [Embodied AI][Multimodal AI][Spatial Understanding]
EMNLP 2025
Yue Zhang, Tianyi Ma, **Zun Wang**, Yanyuan Qiao, Parisa Kordjamshidi

17. **SAME: Learning Generic Language-Guided Visual Navigation with State-Adaptive Mixture of Experts** [Embodied AI]
ICCV 2025
Gengze Zhou, Yicong Hong, **Zun Wang**, Chongyang Zhao, Mohit Bansal, Qi Wu

18. **MVBench: A Comprehensive Multi-Modal Video Understanding Benchmark** [Multimodal AI]
CVPR 2024 (***Highlight***)
Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, **Zun Wang**, Jilan Xu, Guo Chen, Ping Luo, Limin Wang, Yu Qiao

19. **InternVideo2: Scaling Video Foundation Models for Multimodal Video Understanding** [Multimodal AI]
ECCV 2024
Yi Wang*, Kunchang Li*, Xinhao Li*, Jiashuo Yu*, Yinan He*, Guo Chen, Baoqi Pei, Rongkun Zheng, Jilan Xu, **Zun Wang**, Yansong Shi, Tianxiang Jiang, Songze Li, Hongjie Zhang, Yifei Huang, Yu Qiao, Yali Wang, Limin Wang

20. **NavGPT-2: Unleashing Navigational Reasoning Capability for Large Vision-Language Models** [Embodied AI][Multimodal AI]
ECCV 2024
Gengze Zhou, Yicong Hong, **Zun Wang**, Xin Eric Wang, Qi Wu

21. **ETPNav: Evolving Topological Planning for Vision-Language Navigation in Continuous Environments** [Embodied AI]
TPAMI 2024
Dong An, Hanqing Wang, Wenguan Wang, **Zun Wang**, Yan Huang, Keji He, Liang Wang

22. **InternVideo: General Video Foundation Models via Generative and Discriminative Learning** [Multimodal AI]
Technical Report, 2022
Yi Wang*, Kunchang Li*, Yizhuo Li*, Yinan He*, Bingkun Huang*, Zhiyu Zhao*, Hongjie Zhang*, Jilan Xu, Yi Liu, **Zun Wang**, Sen Xing, Guo Chen, Junting Pan, Yali Wang, Limin Wang, Yu Qiao

23. **InternVideo-Ego4D: A Pack of Champion Solutions to Ego4D Challenges** [Multimodal AI]
Technical Report, 2022
Guo Chen, Sen Xing, Zhe Chen, Yi Wang, Kunchang Li, Yizhuo Li, Yi Liu, Jiahao Wang, Yin-Dong Zheng, Bingkun Huang, Zhiyu Zhao, Junting Pan, Yifei Huang, **Zun Wang**, Jiashuo Yu, Yinan He, Hongjie Zhang, Tong Lu, Yali Wang, Limin Wang, Yu Qiao

References

Mohit Bansal — University of North Carolina, Chapel Hill
John R. & Louise S. Parker Distinguished Professor — *mbansal@cs.unc.edu*

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